Draw calls: the terminal and the airport site

Airline Manager Pro · Babylon.js edition · how to draw far less without losing any detail · November 2026

> **The goal.** Cut the number of draw calls in the terminal and at the airport site from about **900–1,000** to about **250–400** inside the building, and make the shadow and glow passes much lighter. The terminal's shape and detail are not touched, and nor is the quality of the graphics: every model, texture, light and effect stays as it is. Only the way the work is handed to the graphics chip changes.

1. What a draw call is, and why it matters here

A draw call is one request from the game to the graphics chip: "draw this piece, with this material". Each one costs time on the processor (preparing the state, the material, the uniforms) before the graphics chip draws anything. With about a thousand of them a frame, the processor becomes the bottleneck, especially on phones and tablets, while the graphics chip waits.

Where the draw calls in the terminal come from today:

Source	Why it costs	Rough share
Structure merged per room and per material (floors, walls, ceilings, cladding)	56 rooms × several materials each	25–30%
Glass walls, each section its own piece	Kept separate to stop it flickering; hundreds of pieces	20–30%
Signs, screens, pictures	Each has its own texture, so its own material and its own draw call	10–15%
Furniture parts (instances)	Cheap: one draw call per shared shape, but many different shapes	10–15%
Aircraft, jet bridges, doors, lifts, the museum, the rental centre	Many separate parts and materials	15–20%
Shadow pass and glow pass	The building is drawn again for the shadows and again for the glow	Up to +100% on top

2. The improvements, biggest first

2.1 Draw only the rooms that can be seen (cells and portals)

What: Today every room on both floors, the pier and the rental centre is drawn whenever it is inside the camera's view, even behind solid walls. The plan already knows every room and every door or opening between them. From that, each frame works out which rooms can be seen from where you stand: your room, the rooms through the openings you can see, and so on, narrowing the view through each opening. Everything else is skipped.

Saves: 50–70% of the draw calls inside the building (from gate 4, the Grand Hall, the offices and the museum can't be seen at all).

Quality: No visible difference: only what can't be seen is skipped. Rooms behind glass walls count as seen.

How: One visibility group per room (its merged structure, its glass, its furniture, its signs). Each door or opening is a portal rectangle. Each frame, flood outwards from the camera's room through the portals that are inside the view, and enable only the groups reached. Outside the building, the glass fronts are portals onto the rooms behind them.

Risk: Low. If the camera is outside every room, everything outside is drawn as today.

2.2 Combine the glass

What: Glass of the same kind on the same floor becomes a handful of pieces instead of hundreds. Flickering is stopped by drawing the glass's depth first (a depth pre-pass) and keeping the glass drawn after everything solid.

Saves: A few hundred draw calls.

Quality: Identical glass: same material, same reflections, same see-through.

Risk: Low. If any glass sorts badly behind other glass, that one facade can be kept as its own piece.

2.3 Shadows drawn once, not every frame

What: The building never moves, but its shadow map is drawn again every frame. It can be drawn once and redrawn only when the sun moves (the time slider, the time-lapse), or a few times a minute at 1×. Moving things (doors, the people later on) get a small separate shadow map of their own.

Saves: The whole shadow pass on most frames: up to half of all draw calls when shadows are on.

Quality: Same shadows, same sharpness.

2.4 Glow only for what glows

What: The glow effect draws the whole scene again, in black where nothing glows. Limited to the meshes that actually glow (the lamps, screens, signs, light strips), it draws a small fraction of the scene.

Saves: Most of the glow pass (at the airport site the ground and sea are already left out).

Quality: Same glow. Something in front of a glowing lamp is still handled by the depth of the scene.

2.5 Signs and screens on one shared texture (an atlas)

What: GATE 4, DEPARTURES, the rental companies, the screen faces and the pictures are painted side by side onto a few large textures. Every sign then shares one material and can be merged per floor.

Saves: 60–120 draw calls.

Quality: The same pixels per metre, so they're just as sharp. Screens that change (the flight boards) keep a separate texture that is updated, still one per floor.

2.6 Merge per floor, keep editing per room

What: The structure is merged per floor and per material instead of per room. Every room and every object keeps its id, its name and its data. The pieces are only combined at load time, and when one room is changed, only its floor is merged again.

Saves: 100–200 draw calls. Works with 2.1 by merging per visibility group (a few rooms at a time).

Quality: No change.

2.7 Cheaper draw calls on the processor

- **Frozen materials:** once built, materials are locked (`material.freeze()`), so they are not checked again every frame. A setting that changes is applied by unlocking, changing and locking again.
- **Frozen active meshes:** in a room where nothing moves, the list of what to draw is kept from frame to frame (`scene.freezeActiveMeshes()`) and only rebuilt when you move to another room or something moves.
- **Thin instances** for the repeated furniture (seats, bins, lamps, planters): the same single draw call per shape as today, with much less work per copy on the processor.
- **Far detail hidden:** very small things (cups, keyboards, labels) are not drawn beyond a few tens of metres, where they are smaller than a pixel.

2.8 Later: WebGPU and recorded frames

On devices that have WebGPU (most new Android phones and PCs), Babylon.js can record the whole frame once and replay it each frame (snapshot rendering). The cost per draw call on the processor then almost disappears. This would be an option in Graphics settings, with WebGL as it is today for every other device.

3. Expected result

Step	Draw calls inside the terminal (about)	Effort	Risk
Today	900–1,000 (+ shadow and glow passes)	–	–
2.1 Visible rooms only	350–500	Medium	Low
2.2 Combined glass	250–400	Small	Low
2.3 + 2.4 Shadows once, glow only what glows	Shadow and glow passes mostly gone	Small	Low
2.5 Atlas for signs	200–330	Medium	Low
2.6 Merge per floor (per visibility group)	150–280	Medium	Low
2.7 Frozen materials, thin instances	Same count, each much cheaper	Small	Low

4. The order to build it in

1. 2.1 visible rooms only, 2.2 combined glass, 2.3 shadows once, 2.4 glow only what glows: the most for the least risk.
2. 2.5 the atlas for signs and screens, 2.6 merging per visibility group.
3. 2.7 the processor savings, then 2.8 WebGPU as a Graphics option.

Each step is released on its own, so the frame counter (FPS, CPU, draw calls, triangles) shows what it changed on the phone and the PC.

5. What is not changed

- The terminal's shape, rooms, walls, glass, roofs, stairs and furniture, and every object's id and name.
- Textures, materials, lights, reflections, shadows' sharpness, glow, bloom and tone mapping.
- The airport site's terrain detail and the weather and sky.